



MedAgent: Trustworthy Personalized Consumer Health Search via Multi-Agent Retrieval and Verification

Ashleyn Castelino, Blazej Madrzyk, Keshav Trikha, Aram Bahrini

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Agenda

- Problem & Background
- Solution Overview
- Demo
- Contributions
- Results
- Next Steps

Problem / Related Work



- **Generic Retrieval:** Existing systems rely on keyword matching or generic embeddings that do not exploit biomedical structure or patient-specific context. [7]
- **Limited Personalization:** Most consumer health systems provide responses with limited adaptation to individual medical profiles or health literacy differences.
- **Weak Trust Controls:** Current LLM systems struggle with uncertainty modeling, evidence weighting, and trustworthy response verification.
- **Gap:** Few prior approaches jointly address retrieval grounding, personalization, and trust in a unified agentic framework.

58.5%

of U.S. adults
search health
info online [1], [3]

88%

have less than
proficient health
literacy [2]

\$106–238B

annual U.S.
healthcare cost
from low literacy [4]

LLM Risks & RAG Accuracy Gains



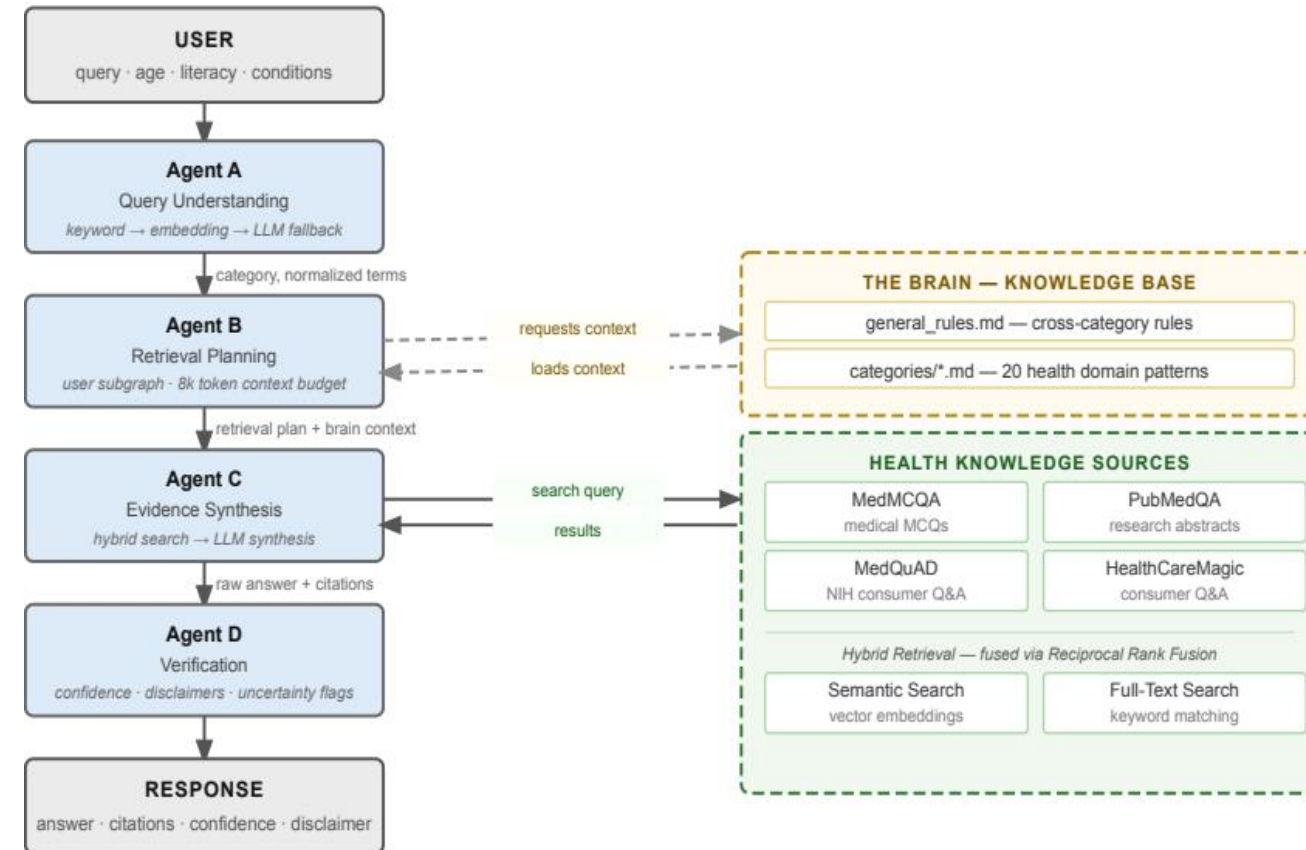
Hallucination rates 50–82% in CDS · RAG lifts accuracy by ~6.6 pp

[9], [10]

Meet MedAgent



- MedAgent: **an agentic consumer health** information retrieval system
- Coordinated multi-agent pipeline decomposes search into four specialized stages
 - **Agent A — Query Understanding:** classifies query into 1 of 20 health categories and expands with HyDE [20]
 - **Agent B — Retrieval Planning:** builds a personalized user subgraph and selects sources
 - **Agent C — Evidence Synthesis:** hybrid search + literacy-tailored LLM synthesis
 - **Agent D — Verification:** confidence scoring, disclaimers, and uncertainty flags
- Shared QueryContext object passed sequentially preserves coherence across stages



MedAgent Principles



01

Multi-Agent Decomposition

Four specialized agents handle intent, planning, synthesis, and verification — interpretable and modular.

02

Tiered Source Strategy

Distinguishes authoritative literature from community Q&A; **weights evidence and surfaces uncertainty.**

03

Personal Health Knowledge Graphs

Per-user subgraph from history drives context-aware retrieval and reranking.



Demo





Contributions

The Brain



- A **structured medical knowledge** base spanning **20 consumer-facing health categories**
- Each category is a markdown file capturing:
 - Terminology map (consumer ↔ clinical translations)
 - Source priority overrides and category-specific retrieval/verification rules
 - Common query patterns, known pitfalls, and links to related categories
- A separate rules **file defines system-wide policies: 4-tier source trust hierarchy, confidence thresholds, multi-source requirements**
- Two roles: (1) grounds Agent A's terminology matching; (2) provides Agent B a token-budgeted context block that shapes Agent C's synthesis prompt

20 Consumer Health Categories

Cardio-vascular	Endocrine / Diabetes	Respiratory	Mental Health	Musculo-skeletal
GI / Digestive	Neuro-logical	Derma-tology	Infectious Disease	Renal / Urinary
Reproductive Health	Pediatrics	Geriatrics	Oncology	Immune / Allergy
Eye / Vision	ENT	Sleep	Nutrition	Medications & Drug Safety

Tiered Source Trust Hierarchy

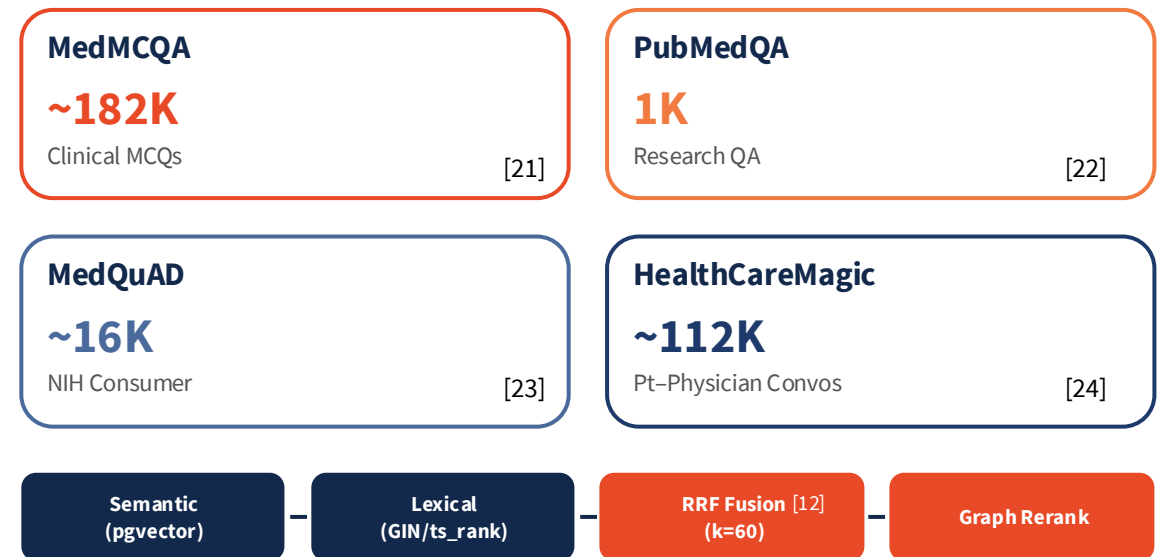
T1 Peer-Reviewed	T2 Clinical QA	T3 NIH Consumer	T4 Forum Q&A
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Tiered Data Retrieval



- Hybrid retrieval over four PostgreSQL-backed biomedical datasets:
 - MedMCQA** — ~182K clinical multiple-choice questions and explanations
 - PubMedQA** — 1K biomedical research QA pairs
 - MedQuAD** — ~16K consumer-facing NIH QA pairs
 - HealthCareMagic** — ~112K real-world consumer–physician conversations
- Semantic search via pgvector + IVFFlat (top-20 per source); lexical search via tsvector + GIN with ts_rank
- Ranked lists fused with **Reciprocal Rank Fusion (k=60)**, then reranked using **personalized graph-derived category weights**
- Higher-authority **sources are prioritized**; uncertainty is surfaced when **sources conflict**

Knowledge Sources



Personalized Knowledge Graph

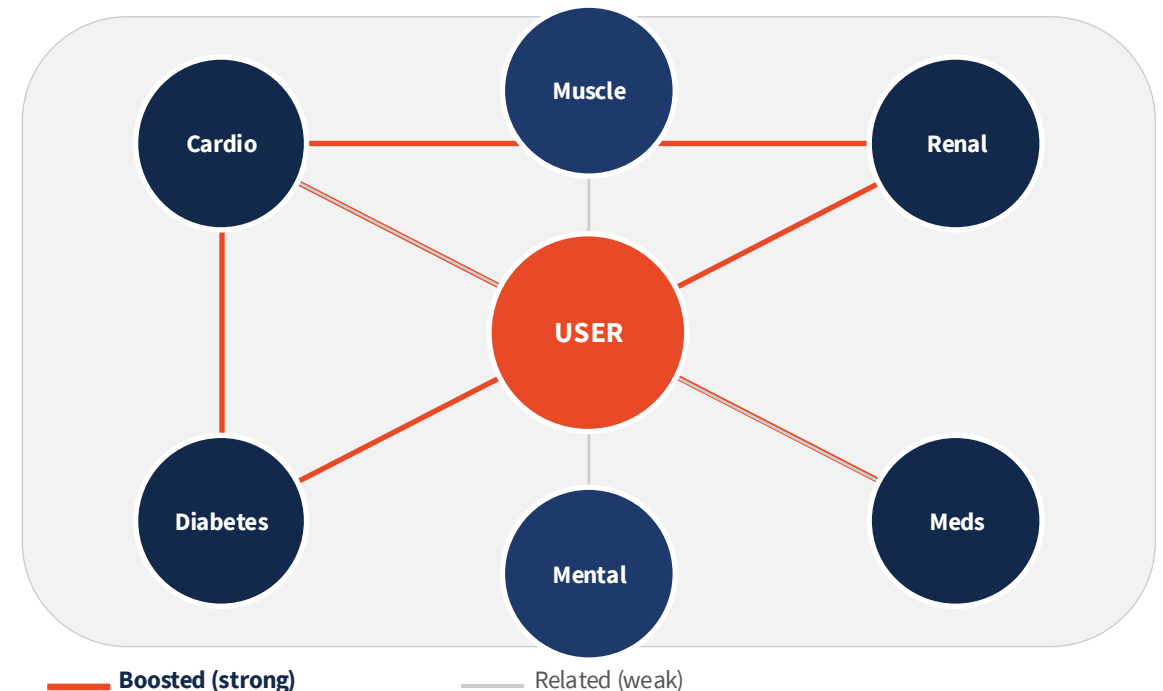


- Base graph: **20 consumer health categories** as nodes with strong/weak/related typed edges

Agent B constructs a **per-user subgraph** through a 5-stage overlay:

1. Anchor active medical conditions to subcategory nodes
 2. Expand related nodes from the anchors
 3. Apply known comorbidity boosts (e.g., diabetes + CKD → cardiovascular)
 4. Apply demographic adjustments (e.g., age 50+ → cardiovascular, musculoskeletal)
 5. Boost medications/drug-safety category and related condition categories from active meds
- Subgraph is encoded into a **Retrieval Plan** that drives source priority and post-fusion reranking (boost factor 0.3, must-load bonus +0.10)

User Subgraph Overlay



Results / Evaluation



HealthBench Performance

- Evaluated on 200 HealthBench [25] examples
 - 5,000 physician-authored rubrics, GPT-4o judge
- MedAgent scores **0.468 vs GPT-4o 0.320 (+46%)**
- Wins 6 of 7 themes — context seeking +71%, global health +54%, health data +50%
- Communication +49%, hedging +47%, emergency referrals +29%; complex responses -1%

HealthBench Scores by Theme

Theme	GPT-4o	MedAgent	Δ
Overall	0.320	0.468	+46%
Communication	0.404	0.602	+49%
Emergency referrals	0.463	0.597	+29%
Hedging	0.354	0.520	+47%
Health data tasks	0.318	0.477	+50%
Context seeking	0.205	0.351	+71%
Complex responses	0.347	0.342	-1%
Global health	0.210	0.323	+54%

Future Work



Modeling & Synthesis

- Strengthen **multi-step clinical reasoning** — the one theme where MedAgent did not beat baseline (complex responses -1%)
- Improve completeness and context awareness via deeper retrieval and broader evidence coverage
- Move from retrieval-based to answer-level confidence that incorporates generation quality

Evaluation & Deployment

- Run user studies of health-literacy adaptation to validate readability gains beyond Flesch-Kincaid
- Tighter integration with clinical-grade knowledge sources and lower-latency deployment
- Broader benchmark coverage beyond the 200-query HealthBench sample
- Develop a self-learning system that learns from mistakes and improves over time



Thank You